# Graphics Programming

# Scalars, Points, Vectors

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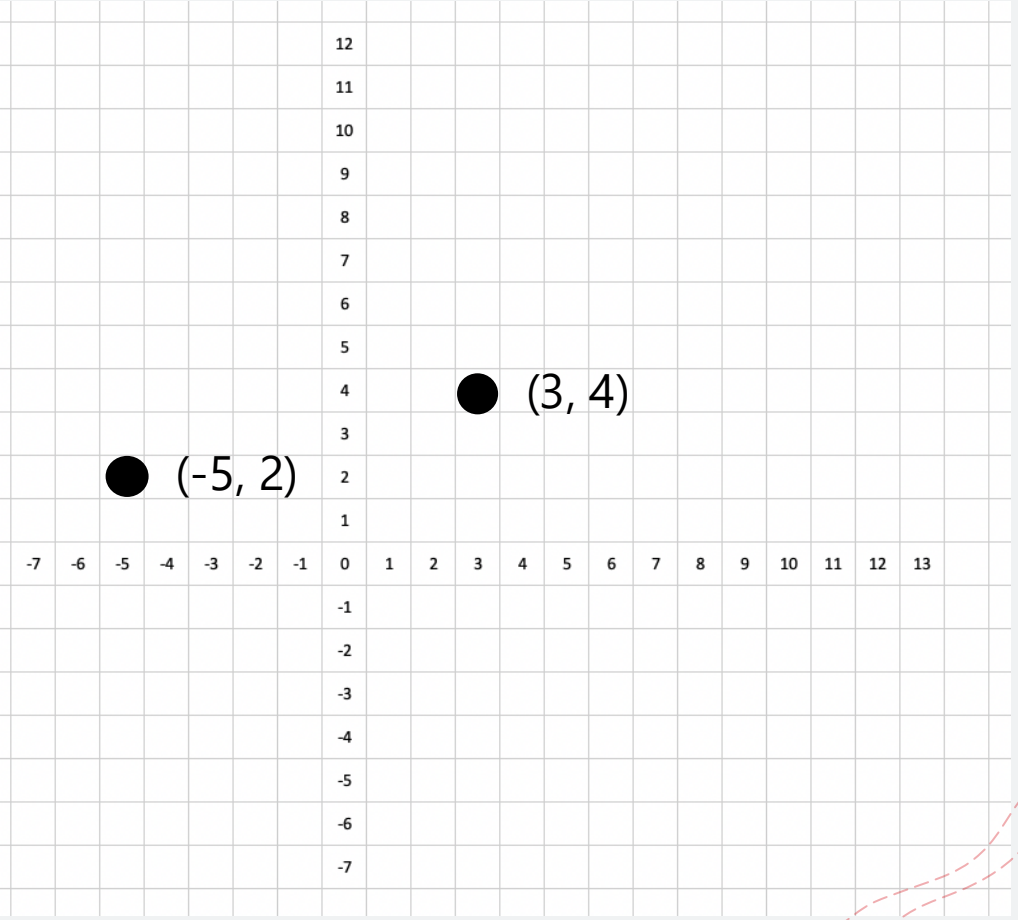
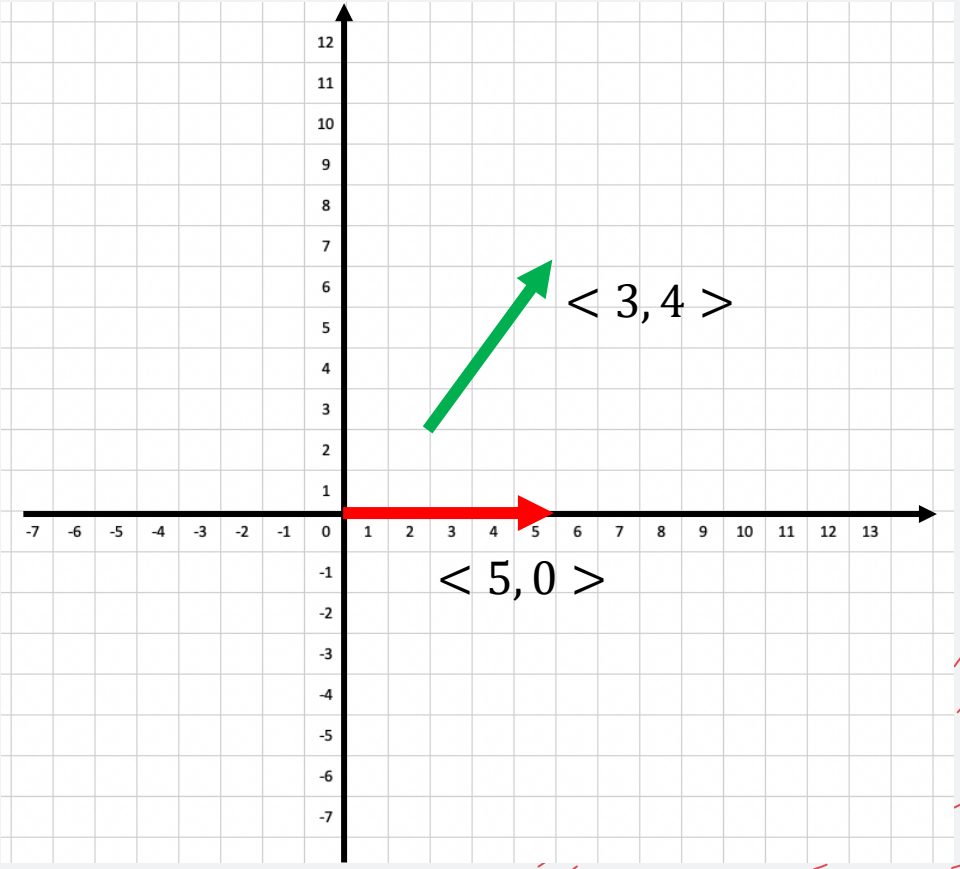
# Point

- A point refers to a location in space
  - E.g., P = (x, y)
- Specified relative to some coordinate system (an origin and axes)
- Typically drawn as a dot



# Vectors

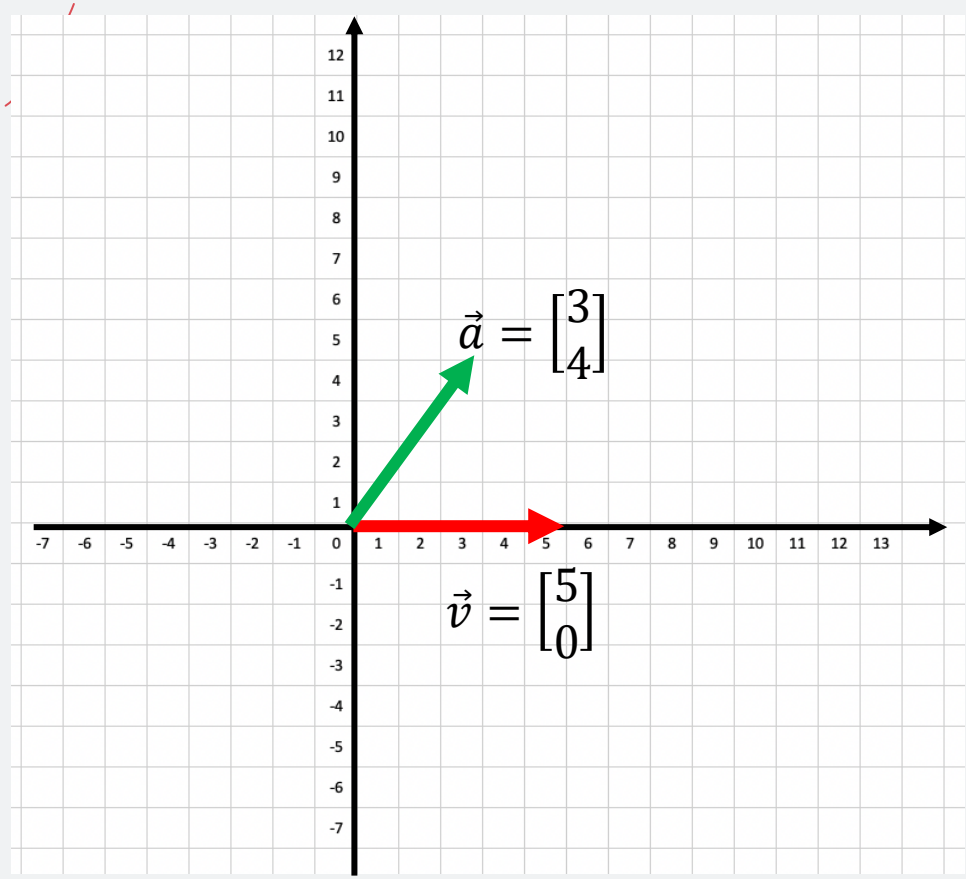
- A vector refers to a *displacement*:
  - An amount of change in each coordinate
  - E.g., v = <m, n>
- Typically drawn as an arrow pointing along the direction of displacement
- Arrow begins at *initial point* or *tail*; ends at *terminal point* or *head*



# Vectors (cont.)

- A vector has both a magnitude and a direction.
  - E.g., An object is moving 5 mph
    - This is not a vector quantity b/c it only has the magnitude
      - This is a scalar quantity
  - E.g., An object is moving 5 mph east
    - This is a vector quantity b/c it has a magnitude and a direction

# Vectors (cont.)




- Rotate a vector will make it to a new vector
- $\vec{a}$ is another vector, which moves 3 in the horizontal direction, and 4 in the vertical direction.
- You can use the Pythagorean theorem to figure out the magnitude of $\vec{a}$
  - $\left|\left|\vec{a}\right|\right| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$

# Points vs Vectors

- Both points and vectors represented by a list of numbers
- Different geometric interpretations
  - Ask a person to meet us at the intersection of 3rd Ave and 5th Street
    - We are talking about a location (point)
  - Ask a person to travel 3 blocks north, 5 blocks east from their current position
    - We are talking about a displacement (a change in position, vector)

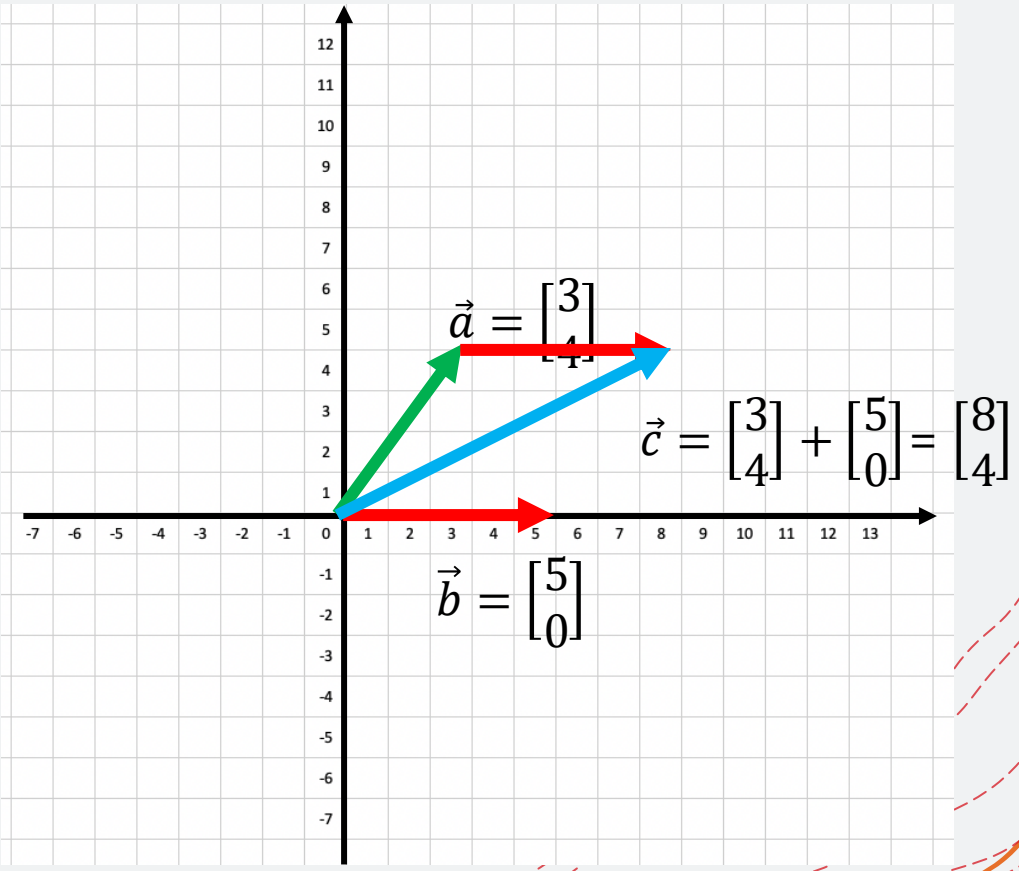
# Representations

- Vectors:
  - Bold lowercase letters, angle brackets $\mathbf{v} = \langle 1, 4 \rangle$
  - Lowercase with an error, square brackets $\vec{v} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$
- Points:
  - Regular uppercase letters, paratheses $P = (2,3)$
- Scalars:
  - Numbers, regular lowercase letters $x = 5$
- Subscripted variables sometimes used when writing components
  - $P = (p_1, p_2)$ $\mathbf{v} = \langle v_1, v_2 \rangle$

# Algebraic Operations

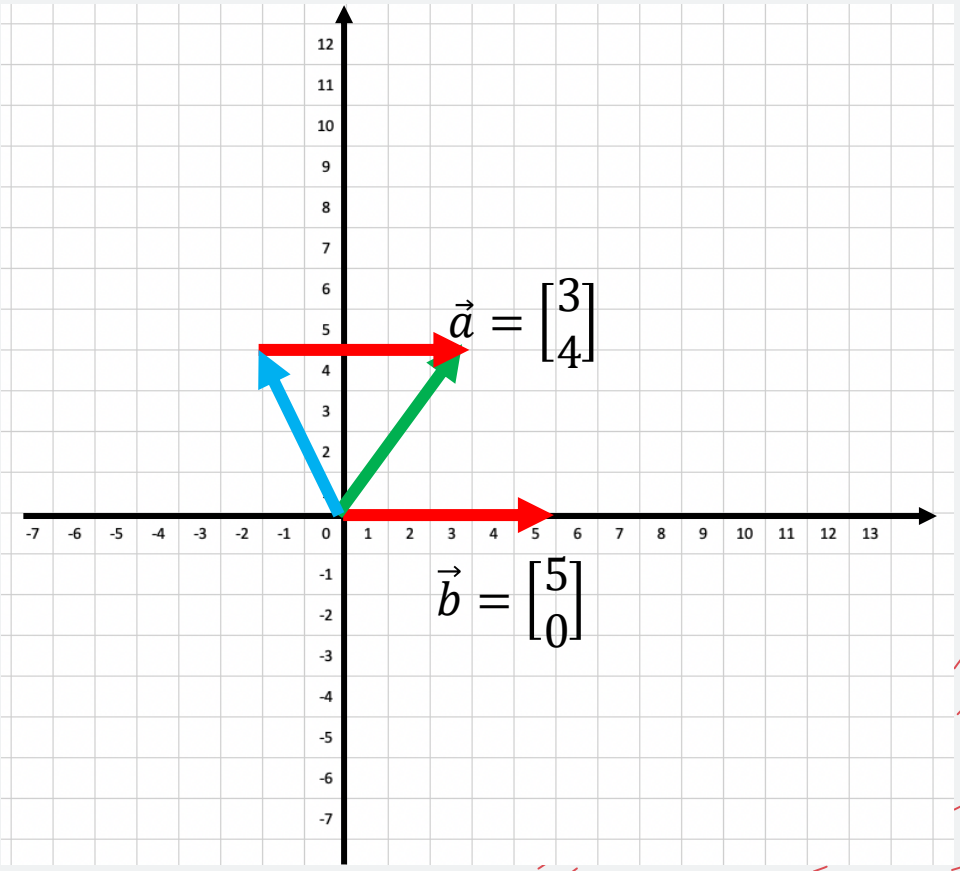
# Vector Addition

- When adding two vectors $\vec{a}$ and $\vec{b}$ together, you obtain a third vector $\vec{c}$ , which contains the addition of each of the vector components.
  - Vector addition, like normal algebraic addition, is commutative meaning that A + B is equal to B + A.
- Geometrically vector addition is described as moving the tail of the $\vec{b}$ vector, which is at the origin, to the head of the $\vec{a}$ vector, which is the value the $\vec{a}$ vector represents in space.
  - The resulting vector $\vec{c}$ is the vector from the tail of A, which is the origin, to the head of B.



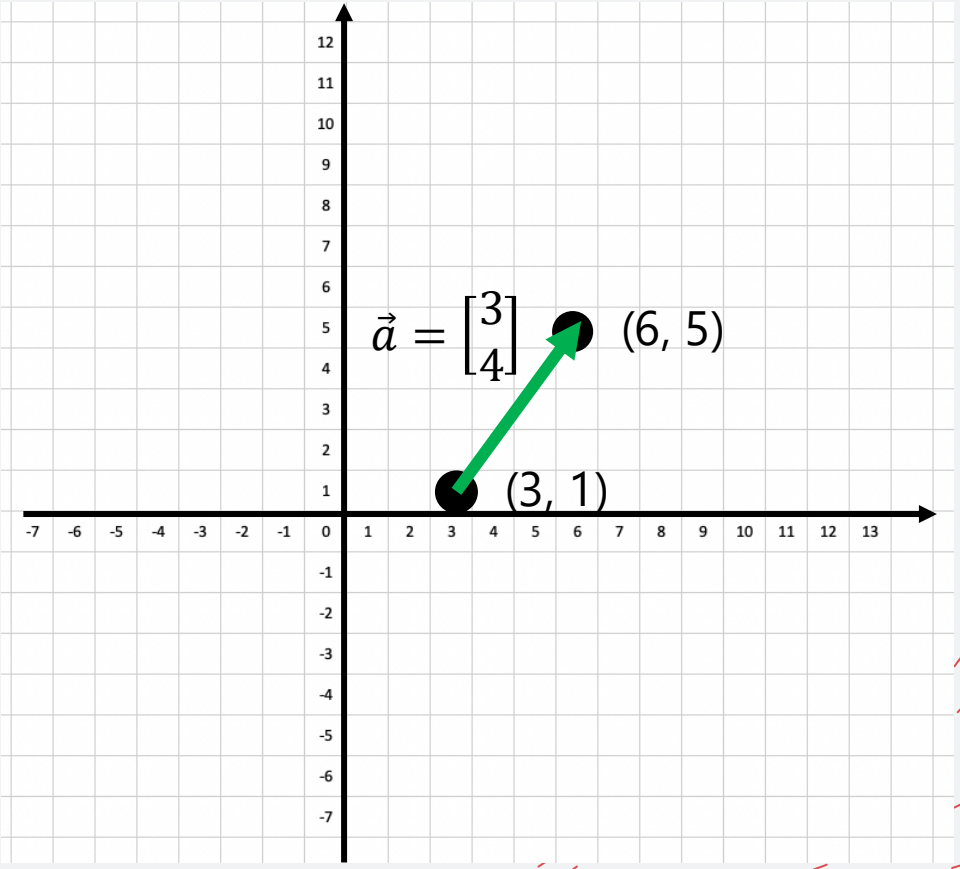
# Vector Subtraction

- When subtracting two vectors $\vec{a}$ and $\vec{b}$ from each other, you obtain a third vector $\vec{c}$, which is the difference of each of the vector components.
  - Vector subtraction is not commutative
    - $\vec{a}$ - $\vec{b}$ is not equal to $\vec{b}$ - $\vec{a}$ .
- Geometrically vector subtraction of $\vec{a}$ - $\vec{b}$ is described as moving the head of vector B to the head of the A vector.
  - The resulting vector $\vec{c}$ is formed from the tail of $\vec{a}$ , which is at the origin, to the tail of $\vec{b}$ .



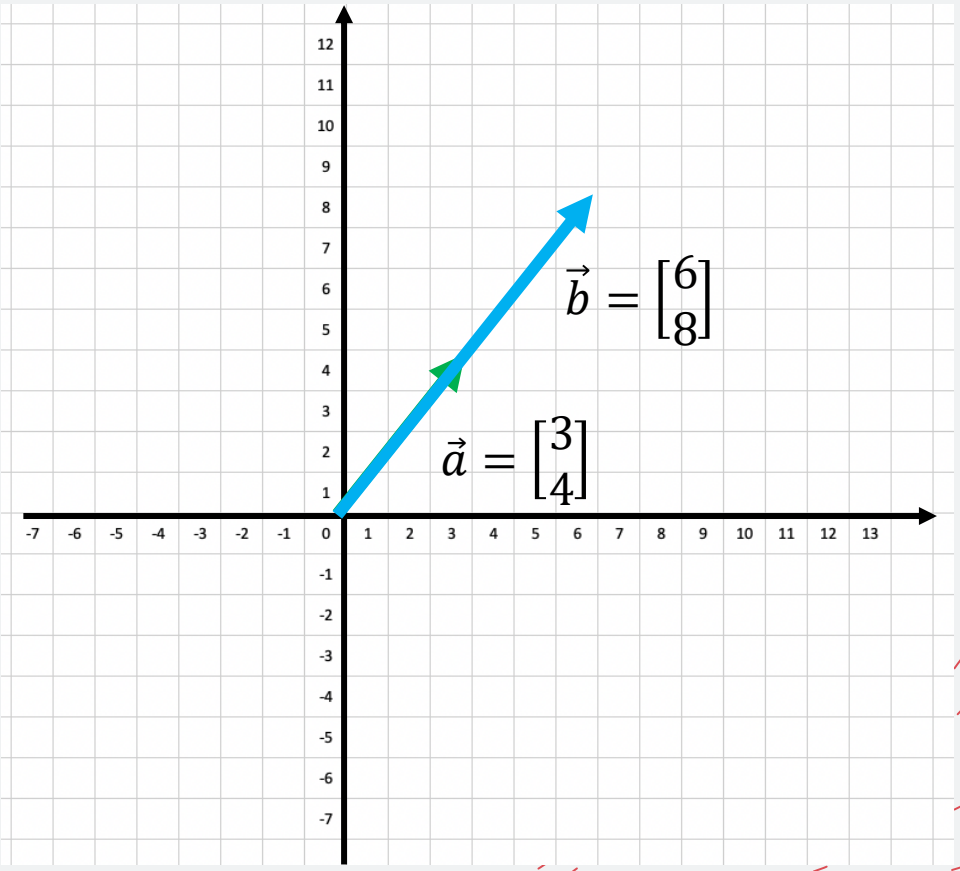
# Vectors Points Addition

- Add vector to point:
  - Translates a vector to a new location
  - Yielding a new point.
- P+ $\vec{a}$ = Q
  - E.g., (3,1)+<3,4> = (6,5)
- Alternatively, $\vec{a}$ = Q-P
  - Geometrically, adding two points does not make sense, but subtracting two points does have a geometrical meaning.
    - Subtracting two points will give us a displament vector



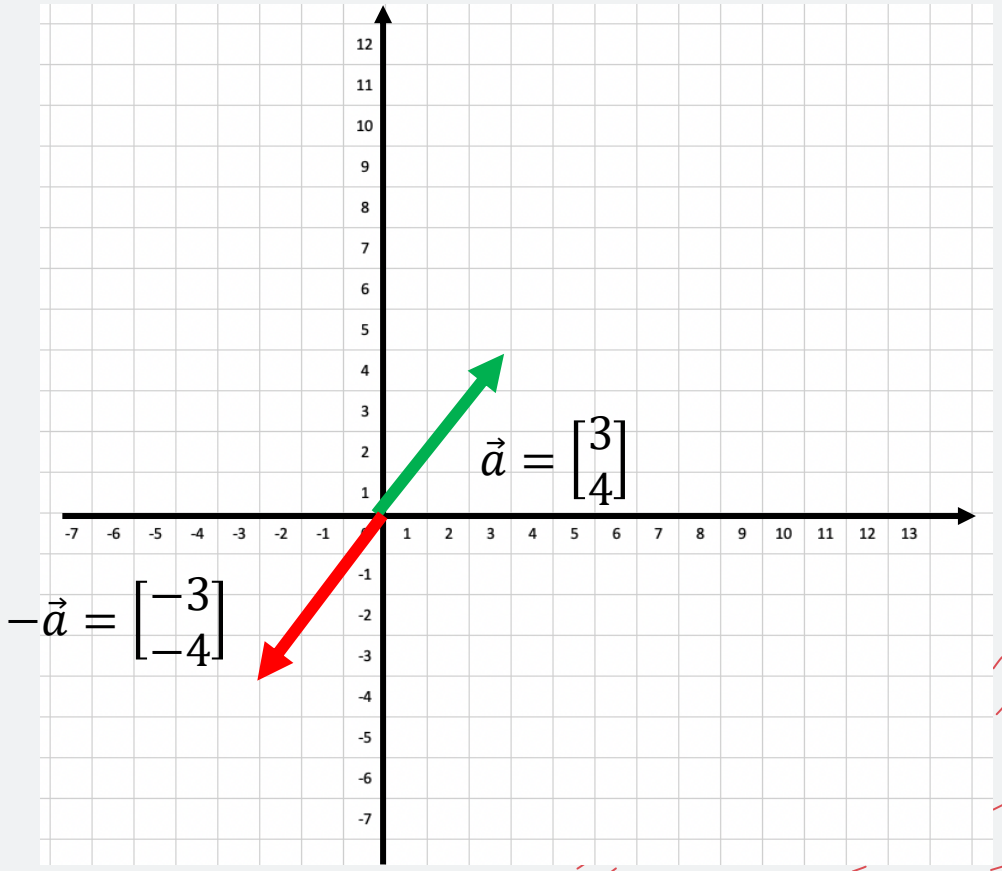
# Vector Scalar Multiplication

- Vectors can be multiplied by a single number called a scalar.
- This scalar value is multiplied against each component to produce a new vector with the same direction but whose length has been multiplied by the scalar.
- Geometrically the multiplication of a vector by a scalar can be described as taking the original vector and stretching or shrinking the length by the scalar value.
  - E.g., $\vec{a} \times 2$



# Vector Negation

- To negate a vector, each of the component's signs are changed to the opposite.
  - Positive components become negative and negative components become positive.
    - This is the same as multiplying the vector by a value of negative one.
  - The negation of a vector is often written $-\vec{a}$, where $\vec{a}$ is the vector name.
- Geometrically the negation of a vector represents a vector in the opposite direction with the same length.



# Basis Vectors

- Consider the vectors
  - **i** = <1, 0> and **j** = <0, 1>
  - Any vector **v** = <x, y> can be written in terms of **i** and **j** using vector multiplication and scalar multiplication in exactly one way:

    **v** = <x,y> = <x,0>+<0,y> = x•<1, 0> + y•<0,1> = x•**i** + y•**j**
- Any set of vectors with these properties is called a *basis*
  - There are many sets of basis vectors
    - **i** and **j** are called the *standard basis*

# Graphics Programming

# Matrices

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# Matrix – In Mathematics

- A matrix is rectangle group of numbers called entries.
- The size of the matrix is expressed in the number of rows by the number of columns.

3 columns

$$A = \begin{bmatrix} -2 & 5 & 6 \\ 5 & 2 & 7 \end{bmatrix}$$

2 rows

2X3 matrix

$$B = \begin{bmatrix} -8 & -4 \\ 23 & 12 \\ 18 & 10 \end{bmatrix}$$

3X2 matrix

# Matrix – In Mathematics (cont.)

- A matrix element is simply a matrix entry.
- Each element in a matrix is identified by naming the row and column in which it appears.
- E.g.,

$$G = \begin{bmatrix} 4 & 14 & -7 \\ 18 & 5 & 13 \\ -20 & 4 & 22 \end{bmatrix}$$

The element $g_{2,1}$ is the entry in the 2nd row and 1st column

$$G = \begin{bmatrix} 4 & 14 & -7 \\ 18 & 5 & 13 \\ -20 & 4 & 22 \end{bmatrix}$$

# Matrix – In Computer Graphics

- The most common type of matrix is the 4 by 4 matrix, which contains 16 float values.
- The matrix is used to represent linear transformations, include
  - Translation, rotation, scaling, shearing, and projection.
- The elements of the matrix create their own coordinate spaces by defining the direction of the X, Y, and Z directions along with the translation from the origin.
- Using a matrix that is 4 by 4 in size enables all the linear transforms to be combined into a single matrix that is used to transform position and direction vectors.

# Matrices Addition

Given $A = \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix}$, let's find $A + B$.

We can find the sum simply by adding the corresponding entries in matrices $A$ and $B$. This is shown below.

$$A + B = \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 4+1 & 8+0 \\ 3+5 & 7+2 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ 8 & 9 \end{bmatrix}$$

# Matrices Subtracting

Similarly, to subtract matrices, we subtract the corresponding entries.

For example, let's consider $C = \begin{bmatrix} 2 & 8 \\ 0 & 9 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 6 \\ 11 & 3 \end{bmatrix}$.

We can find $C - D$ by subtracting the corresponding entries in matrices $C$ and $D$. This is shown below.

$$C - D = \begin{bmatrix} 2 & 8 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 5 & 6 \\ 11 & 3 \end{bmatrix} = \begin{bmatrix} 2-5 & 8-6 \\ 0-11 & 9-3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ -11 & 6 \end{bmatrix}$$

# Matrices Equation

A **matrix equation** is simply an equation in which the variable stands for a matrix.

For example, the equation below is a matrix equation.

$$A + \begin{bmatrix} 3 & 5 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 10 & 12 \end{bmatrix}$$

A variable substitution makes solving this matrix equation easy.

If we let $B = \begin{bmatrix} 3 & 5 \\ 2 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ 10 & 12 \end{bmatrix}$, we obtain this equation:

$$A + B = C$$

$$A = C - B \qquad \text{Subtract B}$$

We can now substitute matrices $B$ and $C$ and solve for $A$.

$$A = C - B$$

$$= \begin{bmatrix} 1 & 0 \\ 10 & 12 \end{bmatrix} - \begin{bmatrix} 3 & 5 \\ 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1-3 & 0-5 \\ 10-2 & 12-2 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -5 \\ 8 & 10 \end{bmatrix}$$

# Multiplying Matrices by Scalars

The term scalar multiplication refers to the product of a real number and a matrix. In scalar multiplication, each entry in the matrix is multiplied by the given scalar.

$$2A = 2 \cdot \begin{bmatrix} 10 & 6 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 \cdot 10 & 2 \cdot 6 \\ 2 \cdot 4 & 2 \cdot 3 \end{bmatrix}$$

$$= \begin{bmatrix} 20 & 12 \\ 8 & 6 \end{bmatrix}$$

# Matrix Multiplication

- Product of two matrices **A** and **B**
- Number of columns of **A** must equal number of rows of **B**
  - If **A** is a n × m matrix, and **B** is an m × p matrix, then **AB** is an

    n × p matrix
- Entries of the product are given by

$$M_{ij} = (AB)_{ij} = \sum_{k=1}^{m} A_{ik} B_{kj}$$

# Matrix Multiplication

Example matrix product:

$$M = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 4 & -5 \end{bmatrix} = \begin{bmatrix} 8 & -13 \\ -6 & 6 \end{bmatrix}$$

$M_{11}$ = 2*(-2) + 3*4 = 8

$M_{12}$ = 2*1 + 3*(-5) = -13

$M_{21}$ = 1*(-2) + (-1)*4 = -6

$M_{22}$ = 1*1 + (-1) *(-5) = 6

$$\boxed{M_{ij} = (AB)_{ij} = \sum_{k=1}^{m} A_{ik} B_{kj}}$$

# Vectors and Matrices Multiplication

- Matrices are used to transform vectors from one coordinate system to another.
- In three dimensions, the product of a row vector and a matrix looks like:

$$[V_x \quad V_y \quad V_z]\begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix} = [R_1 \quad R_2 \quad R_3]$$

$$R_1 = V_x * M_{11} + V_y * M_{21} + V_z * M_{31}$$
$$R_2 = V_x * M_{12} + V_y * M_{22} + V_z * M_{32}$$
$$R_3 = V_x * M_{13} + V_y * M_{23} + V_z * M_{33}$$

$$M_{ij} = (AB)_{ij} = \sum_{k=1}^{m} A_{ik} B_{kj}$$